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Assignment II
SDG-Based Questions – CO2

Course Code ITA06
Course Name Machine Learning

concept to a real-world application aligned with the United Nations Sustainable Development Goals (SDGs).

Discuss the role of convolutional and pooling layers (conceptually) in a deep network designed to classify satellite images of deforested vs forested land. (SDG 15 – Life on Land)

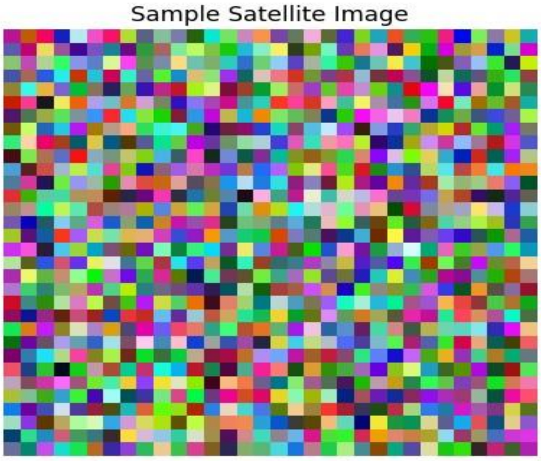
```
import tensorflow as tf
import numpy as np
import matplotlib.pyplot as plt
```

```
# Create 100 random satellite images (32x32 RGB)

X = np.random.rand(100,32,32,3)

# Labels: 0 = Forest, 1 = Deforested
y = np.random.randint(0,2,100)

print(X.shape)
print(y.shape)
```

```
(100, 32, 32,3)
(100,)
```

```
plt.imshow(X[0])
plt.title("Sample Satellite Image")
plt.axis("off")
plt.show()
```

Sample Satellite Image



```
model = tf.keras.Sequential([

    tf.keras.layers.Conv2D(
        16,(3,3),
         activation='relu',
         input_shape=(32,32,3)
    ),

    tf.keras.layers.MaxPooling2D(2,2),

    tf.keras.layers.Flatten(),

    tf.keras.layers.Dense(
        16,
         activation='relu'
    ),

    tf.keras.layers.Dense(
        2,
```

```
        activation='softmax'
    )

])

model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/convolutional/base_conv.py:113: UserWarni
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

```
history = model.fit(
    X,
    y,
    epochs=2,
    batch_size=16
)
```

```
Epoch 1/2
7/7 ━━━━━━━━━━━━━━━━━━━━ 1s 14ms/step - accuracy: 0.5600 - loss: 0.6950
Epoch 2/2
7/7 ━━━━━━━━━━━━━━━━━━━━ 0s 13ms/step - accuracy: 0.6200 - loss: 0.6580
```

```
plt.plot(history.history['accuracy'], marker='o')
plt.title("Training Accuracy")
plt.xlabel("Epoch")
plt.ylabel("Accuracy")
plt.grid(True)
plt.show()
```


```
prediction = model.predict(X[:1])

classes = ["Forest","Deforested"]

print("Prediction:", classes[np.argmax(prediction)])
```

```
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 79ms/step
Prediction: Forest
```